

The complete system the solver solves

Ti-O gas-solid Gibbs equilibrium · solidgas/activeset.py · every output is this one minimisation

1 · Inputs → element inventory

$$n_{g,0} \; = \; \frac{PV}{R_{\text{atm}}\,T_{\text{charge}}}$$

$$y_{i,0} \; = \; \frac{r_i}{\sum_k r^k}, \qquad n_{i,0} \; = \; y_{i,0}\,n_{g,0}$$

$$n_{s,0} \; = \; \frac{n_{\text{solid}}}{t_h W_{\text{Ti}} + o_h W_{\text{O}}}$$

$$b_e \; = \; \sum_i a_{ie}\,n_{i,0} \; + \; a_{he}\,n_{s,0}, \qquad e \in \left\{ \text{C, H, N, O, Ti} \right\}$$

$$i \in \left\{ \text{CO}_2, \; \text{H}_2, \; \text{CO}, \; \text{H}_2\text{O}, \; \text{N}_2, \; \text{O}_2 \right\}; \qquad j \in \left\{ \text{TiO}_2, \; \text{Ti}_{10}\text{O}_{19}, \; \text{Ti}_8\text{O}_{15}, \; \text{Ti}_6\text{O}_{11}, \; \text{Ti}_5\text{O}_9, \; \text{Ti}_4\text{O}_7, \; \text{Ti}_3\text{O}_5, \; \text{Ti}_2\text{O}_3 \right\} = \text{Ti}_{t_j}\text{O}_{o_j}$$

2 · Standard chemical potentials

gases — NIST-JANAF Shomate, $t = T/1000$:

$$H^\circ(T) = \Delta_f H_{298}^\circ + At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H$$

$$S^\circ(T) = A \ln t + Bt + \frac{Ct^2}{2} + \frac{Dt^3}{3} - \frac{E}{2t^2} + G, \qquad \mu^\circ(T) = H^\circ(T) - \frac{TS^\circ(T)}{1000}$$

condensed phases — Waldner & Eriksson CALPHAD, integrated piecewise from 298.15 K :

$$\mu^\circ(T) = \frac{1}{1000} \left[\Delta_f H_{298}^\circ + \sum_{\text{seg}} \left(\Delta H_{tr} + \int C_p \, dT \right) - T \left(S_{298}^\circ + \sum_{\text{seg}} \left(\frac{\Delta H_{tr}}{T_{tr}} + \int \frac{C_p}{T} \, dT \right) \right) \right]$$

$$C_p = a + bT + cT^2 + dT^{-2} \quad (\text{poly}); \qquad C_p = k_0 + k_1T^{-1/2} + k_2T^{-2} + k_3T^{-3} \quad (\text{magnéli})$$

3 · The master problem

$$\min_{n,m} \quad G(n,m) \; = \; \sum_i n_i \Big[\mu_i^\circ + RT \ln \Big(\frac{n_i}{N_g} \cdot \frac{P}{P^\circ} \Big) \Big] \; + \; \sum_j m_j \mu_j^\circ$$

$$\text{s. t.} \quad \sum_i a_{ie}\,n_i + \sum_j a_{je}\,m_j = b_e \quad \forall e, \qquad n_i \geq 0, \quad m_j \geq 0, \qquad N_g = \sum_i n_i$$

ideal gas mixture; each condensed phase a pure line compound, $a_j = 1$; $P^\circ = 1 \text{ atm}$, $R = 8.314472 \times 10^{-3} \text{ kJ mol}^{-1}\text{K}^{-1}$

4 · KKT conditions — solved per candidate active set A

$$n_i > 0: \qquad \mu_i^\circ + RT \ln(y_i P/P^\circ) \; = \; \sum_e a_{ie} \lambda_e$$

$$j \in A: \qquad \mu_j^\circ \; = \; t_j \lambda_{\text{Ti}} + o_j \lambda_{\text{O}}, \qquad m_j > 0$$

$$j \notin A: \qquad m_j = 0, \qquad r_j \equiv \mu_j^\circ - t_j \lambda_{\text{Ti}} - o_j \lambda_{\text{O}} \geq 0$$

$$\sum_i a_{ie}\,y_i N_g + \sum_j a_{je}\,m_j = b_e, \qquad \sum_i y_i = 1$$

5 · Decomposed solve

$|A| = 2 \quad (\text{two-phase buffer} \text{ — the pair pins both solid potentials, linearly}):$

$$t_1 \lambda_{\text{Ti}} + o_1 \lambda_{\text{O}} = \mu_1^\circ \qquad \wedge \qquad t_2 \lambda_{\text{Ti}} + o_2 \lambda_{\text{O}} = \mu_2^\circ$$

gas Newton over $x = (\theta_e, \ln N_g)$, $\theta_e = \lambda_e/RT$:

$$\ln y_i \; = \; \sum_e a_{ie} \theta_e \; - \; \frac{\mu_i^\circ}{RT} \; - \; \ln \frac{P}{P^\circ}$$

$$F_e = \sum_i a_{ie}\,y_i N_g - \hat{b}_e = 0, \qquad F_0 = \sum_i y_i - 1 = 0$$

$$\frac{\partial F_e}{\partial \theta_f} = \sum_i a_{ie} a_{if} y_i N_g, \qquad \frac{\partial F_e}{\partial \ln N_g} = \sum_i a_{ie} y_i N_g, \qquad x \leftarrow x - \alpha J^{-1} F, \quad \alpha: \|F\| \downarrow$$

condensed amounts recovered linearly — the combination $\text{O}' = \text{O} - \rho_h \text{Ti}$ removes the host oxygen symbolically:

$$\sum_j t_j m_j = b_{\text{Ti}}, \qquad \sum_i a_{i\text{O}} y_i N_g = b_{\text{O}}^{\text{gas}} + \sum_j d_j m_j, \qquad d_j \equiv \rho_h t_j - o_j, \quad \rho_h = o_h/t_h$$

$|A| = 1: \qquad m_1 = b_{\text{Ti}}/t_1, \qquad \lambda_{\text{O}} = RT \theta_{\text{O}}, \qquad \lambda_{\text{Ti}} = (\mu_1^\circ - o_1 \lambda_{\text{O}})/t_1$

$|A| = 3 \text{ (probe)}: \qquad \text{res}_3 = \mu_3^\circ - t_3 \lambda_{\text{Ti}} - o_3 \lambda_{\text{O}}; \qquad |\text{res}_3| < \varepsilon_{\text{deg}} \Leftrightarrow \text{invariant point}$

6 · Decision rules

$G \text{ convex} \Rightarrow [m_j > 0 \; \forall j \in A \; \wedge \; r_j \geq 0 \; \forall j \notin A] \Leftrightarrow \text{global minimum}$

$|r_j|/d_j < \varepsilon_{\text{deg}} \Rightarrow \text{degenerate boundary (reported, never resolved arbitrarily)}$

$b_e = 0 \Rightarrow n_i = 0 \; \forall i \ni e \quad (\text{removed from the variables: exactly zero})$

$\text{avail}_{\text{O}} = 0 \; \wedge \; \exists j \notin A: d_j > 0 \Rightarrow \frac{\partial G}{\partial \xi}|_{0^+} \sim \frac{RT}{2} \ln \xi \rightarrow -\infty \Rightarrow \text{reject } A$

7 · Reduced cost, two bases

$r_j^{\text{formula}} = \mu_j^\circ - t_j \lambda_{\text{Ti}} - o_j \lambda_{\text{O}} \quad [\text{kJ/mol compound}], \qquad r_j^{\text{O}} = r_j^{\text{formula}}/d_j \quad [\text{kJ/mol O}]$

every $\text{TiO}_2 \rightarrow$ Magnéli step has $d_j = 1$, so the two bases coincide for these steps

8 · Derived outputs

$$\text{Ti}^{3+} \% = 100 \frac{\sum_j c_j m_j}{b_{\text{Ti}}}, \quad c_j = 2 \; (j \neq \text{TiO}_2); \qquad x = 2 - \frac{\sum_j o_j m_j}{\sum_j t_j m_j}$$